Recapitulare parțial

Recapitulare

1) $\int_{|z|=4} \frac{z\,dz}{\sin z(1-\cos z)}$; teorema reziduurilor

2) $\int_{|z|=4} \frac{dz}{3\sin z - \sin 3z}$; teorema reziduurilor

3) $\int_{|z|=2} \frac{z^n \cdot e^{\frac{1}{z}}}{z^2-1} dz$; teorema reziduurilor

4) $\int_0^{2\pi} \frac{\cos n\theta}{1-2a\cos\theta+a^2} d\theta$; $a>1$

5) $\int_{-\infty}^{\infty} \frac{x\cos x}{x^2-6x+13} dx$;

↳ integrale reale de tipul $\int_{-\infty}^{\infty}$ (prelungești cu $\sin n\theta$)

6) $\int_0^{\infty} \frac{x\sin ax}{(x^2+b^2)(x^2+c^2)} dx$, $b, c > 0$

7) Să se det. transf. Laplace a fct.

$$f(t) = \begin{cases} 2, & 0<t<1 \\ \frac{t^2}{2}, & 1<t<\frac{\pi}{2} \\ \cos t, & t>\frac{\pi}{2} \end{cases}$$

8) $y'' + 3y' + 2y = u(t-1) - u(t-2)$, $y(0)=0$, $y'(0)=0$

9) $y'' + 4y = \begin{cases} 8t^2, & 0<t<5 \\ 200, & t>5 \end{cases}$, $y(1) = 1+\cos 2$; $y'(1) = 4 - 2\sin 2$

10) $x''(t) - 2x'(t-1) = t$, $x(0) = x'(0) = 0$

11) $x(t) = t + 2\int_0^t [t - \tau - \sin(t-\tau)]\, x(\tau)\, d\tau$

→ vezi proprietăți transformata Fourier

12) Det transf. Fourier a semnalului:

$$f(t) = (t+2)\, e^{-(t+2)(3i+5)}, \quad t > 0.$$

13) Rez. ec. integrală (cu transf. Fourier):

$$\int_0^\infty f(t) \cdot \sin(\omega t)\, dt = \begin{cases} \frac{\pi}{2} \sin\omega, & \omega \in (0, \pi) \\ 0, & \omega \geq \pi \end{cases}$$

14) Rez. cu tr. Laplace:

$$y'' + y' - 2y = \begin{cases} 3\sin t - \cos t, & 0 < t < 2\pi \\ 3\sin 2t - \cos 2t, & t > 2\pi. \end{cases}$$

$$y(0) = 1, \quad y'(0) = 0$$

Rezolvări

3. $$\int_{|z|=2} \frac{z^n \cdot e^{\frac{1}{z}}}{z^2 - 1}\, dz$$

$-1, +1$ poli simpli

0 punct singular - -

$$\operatorname{Rez}(f, -1) = \lim_{z \to -1} (z+1) \cdot \frac{z^n \cdot e^{\frac{1}{z}}}{(z+1)(z-1)} = \frac{(-1)^n \cdot e^{-1}}{-2}$$

$$\operatorname{Rez}(f, 1) = \lim_{z \to 1} (z-1) \cdot \frac{z^n \cdot e^{\frac{1}{z}}}{(z+1)(z-1)} = \frac{e}{2}$$

$$\frac{z^n \cdot e^{1/z}}{z^2 - 1} = -z^n \cdot \sum_{k \geq 0} \frac{1}{k!} \left(\frac{1}{z}\right)^k \cdot \sum_{k \geq 0} z^{2k} =$$

$$= -z^n \left(1 + \frac{1}{1!}\frac{1}{z} + \frac{1}{2!}\cdot\frac{1}{z^2} + \ldots + \frac{1}{n!}\cdot\frac{1}{z^n} + \ldots\right)\left(1 + z^2 + z^4 + \ldots + z^{2n}\right) \Rightarrow$$

$z = 0$ este punct singular esențial și

$\text{Rez}(f,0) = -\left(\frac{1}{(n+1)!} + \frac{1}{(n+3)!} + \frac{1}{(n+5)!} + \dots\right)$

$\Rightarrow \int_{|z|=2} f(z)\,dz = 2\pi i \cdot (\text{Rez}(f,-1) + \text{Rez}(f,1) + \text{Rez}(f,0))$

(4.) $i = \int_0^{2\pi} \frac{\cos n\theta}{1 - 2a\cos\theta + a^2}\,d\theta \quad a > 1$

Integrala este de tipul I și facem schimbarea

$z = e^{i\theta}$, $\theta \in [0, 2\pi] \Rightarrow z$ parcurge cercul unitate, adică $|z| = 1$ și $dz = ie^{i\theta}d\theta = iz\,dt$

Considerăm integrala $J = \int_0^{2\pi} \frac{\sin\theta n}{1 - 2a\cos\theta + a^2}\,d\theta = 0$

$= 0$ din imparitatea funcției sin $\Rightarrow$

$$i = i + Ji = \int_0^{2\pi} \frac{\cos n\theta + i\sin n\theta}{1 - 2a\cos\theta + a^2}\,d\theta = \int_0^{2\pi} \frac{e^{in\theta}}{1 - 2a\cos\theta + a^2}\,d\theta =$$

$$= \int_{|z|=1} \frac{z^n}{1 - a(z + \frac{1}{z}) + a^2} \cdot \frac{1}{iz}\,dz = \frac{1}{i}\int_{|z|=1} \frac{z^n}{z - a(z^2+1) + a^2 z}\,dz$$

$$= \frac{-1}{i}\int_{|z|=1} \frac{z^n}{az^2 - z(a^2+1) + a}\,dz$$

$\Delta = (a^2+1)^2 - 4a^2 = a^4 + 2a^2 + 1 - 4a^2 = (a^2-1)^2$

$a_{1,2} = \frac{a^2+1 \pm (a^2-1)}{2a}$ ⓐ $\notin \text{Int}(|z|=1)$

ⓑ $\frac{1}{a} \in \text{Int}(|z|=1)$ pol de ord. 1

$\Rightarrow i = \frac{-1}{i} 2\pi i \cdot \text{Rez}(f, \frac{1}{a}) = -2\pi\,\text{Rez}(f, \frac{1}{a})$

$\frac{1}{a} \to$ pol simplu

$$\Rightarrow \text{Rez}(f, \tfrac{1}{a}) = \lim_{z \to 1/a} (z - \tfrac{1}{a}) \cdot \frac{z^n}{(z - \frac{1}{a})(z-a)} = \frac{(\frac{1}{a})^n}{\frac{1}{a} - a}$$

$$= \frac{\left(\frac{1}{a}\right)^n}{\frac{1-a^2}{a}} = \frac{\left(\frac{1}{a}\right)^{n-1}}{1-a^2}$$

$$\Rightarrow \quad i = i + i\int = -2\pi \cdot \frac{\left(\frac{1}{a}\right)^{n-1}}{1-a^2}$$

(1.) $$i = \int_{|z|=4} \frac{z\,dz}{\sin z(1-\cos z)} = \int_{|z|=4} \frac{z\,dz}{\sin z \cdot 2\sin^2\frac{z}{2}}$$

$$\sin z = 0 \iff \frac{e^{iz} - e^{-iz}}{2i} = 0 \iff e^{iz} - e^{-iz} = 0 \iff$$

$$e^{2iz} = 1 \iff 2iz = \ln 1 \iff 2iz = \ln|1| + i(0 + 2k\pi),\ k \in \mathbb{Z}$$

$$\Rightarrow \cancel{2i}z = \cancel{2k\pi i} \iff z = k\pi \Rightarrow -\pi, 0, \pi \in \operatorname{Int}(|z|=4)$$

$$\cos z = 1 \iff \frac{e^{iz} + e^{-iz}}{2} = 1 \iff e^{iz} + e^{\bar{i}z} = 2 \iff e^{2iz} - 2e^{iz} + 1 = 0$$

$$\iff (e^{iz} - 1)^2 = 0 \iff e^{iz} = 1 \iff \dots \iff z = 2k\pi \Rightarrow$$

$$0 \in \operatorname{Int}(|z|=4)$$

$$\Rightarrow \begin{cases} z = \pm\pi & \text{poli simpli} \\ z = 0 & \text{pol dublu} \end{cases}$$

$$\Rightarrow \int_{|z|=4} \frac{2\,dz}{\sin z(1-\cos z)} = 2\pi i\left(\operatorname{Rez}(f,0) + \operatorname{Rez}(f,\pi) + \operatorname{Rez}(f,-\pi)\right)$$

$$\operatorname{Rez}(f,-\pi) = \lim_{z\to-\pi} \left((z+\pi)\cdot \frac{z}{-\sin(z+\pi)(1-\cos z)}\right) =$$

$$= \frac{-\pi}{+(1-(-1))} = \frac{\pi}{2}$$

$$\operatorname{Rez}(f,\pi) = \lim_{z\to\pi} \left((z-\pi)\cdot \frac{z}{-\sin(z-\pi)(1-\cos z)}\right) = \frac{-\pi}{2}$$

$$\frac{z}{\sin z(1-\cos z)} = z \cdot \frac{1}{\sin z} \cdot \frac{1}{1-\cos z} = z \cdot$$

$$\operatorname{Rez}(f,0) = \lim_{z\to 0}\left[z^2 \cdot \frac{z}{\sin z(1-\cos z)}\right]' \overset{\text{calcule}}{=} \ldots =$$

$$\lim_{z\to 0} \frac{z^2(3\sin z - z(2\cos z+1))}{(\cos(z)-1)\sin^2 z} = \lim_{z\to 0} -\frac{3\sin z - z(2\cos z+1)}{\cos z - 1}$$

$$= \lim_{z\to 0} -\left(\frac{3\sin\frac{z}{2}\cos\frac{z}{2}}{2\sin^2\frac{z}{2}} - \frac{2\cos z+1}{2\sin^2\frac{z}{2}}\right) =$$

$$= \lim_{z\to 0} \frac{1}{\sin\frac{z}{2}}\left(3\cos\frac{z}{2} + 2\cos z + 1\right) =$$

$$\lim_{z\to 0} \frac{2\cos z + 1 - 3\cos\frac{z}{2}}{\sin\frac{z}{2}} = \ldots \text{ calcule}$$

$$\Rightarrow \boxed{I = 2\pi i\,(\operatorname{Rez}(f,0) + \operatorname{Rez}(f,\pi) + \operatorname{Rez}(f,-\pi))}$$

(2.) $$I = \int_{|z|=4} \frac{dz}{3\sin z - \sin 3z} = \int_{|z|=4} \frac{dz}{3\sin z - 3\sin z + 4\sin^3 z}$$

$$= \int_{|z|=4} \frac{dz}{4\sin^3 z} \Rightarrow$$

$\sin z = 0 \iff z = k\pi \Rightarrow z \in \{-\pi, 0, \pi\}$

$$\Rightarrow \begin{cases} z = \pm\pi \\ z = 0 \end{cases}$$ poli tripli

$$\lim_{z\to 0} \frac{1}{2!}\left[z^3 \cdot \frac{1}{4\sin^3 z}\right]'' \overset{\text{calcule}}{=} \ldots = \frac{1}{8} = \operatorname{Rez}(f,0)$$

$$\lim_{z\to \pi} \frac{1}{2!}\left[(z-\pi)^3 \cdot \frac{1}{4\sin^3 z}\right] \overset{\text{calcule}}{=} \ldots = -\frac{1}{4}\cdot\frac{1}{2} = -\frac{1}{8} = \operatorname{Rez}(f,\pi)$$

Analog $\operatorname{Rez}(f, -\pi) - -$

$\Rightarrow$ $i = 2\pi i \left(\operatorname{Rez}(f,0) + \operatorname{Rez}(f,-\pi) + \operatorname{Rez}(f,\pi) \right)$

(12.) Transformata Fourier a semnalului

$f(t) = (t+2)\, e^{-(t+2)(3i+5)}$, $t > 0$

$\mathcal{F}[e^{-t}](\omega) = \frac{1}{1+i\omega}$

$e^{-5t} \xrightarrow{\mathcal{F}} \frac{1}{5+i\omega}$

$t \cdot e^{-5t} \xrightarrow{\mathcal{F}} i \cdot \left(\frac{1}{5+i\omega}\right)' = i \cdot \frac{-i}{(5+i\omega)^2} = \frac{1}{(5+i\omega)^2}$

(derivarea în rap. cu frecvența)

$\underbrace{t \cdot e^{-5t} \cdot e^{-3it}}_{\| \; t \cdot e^{-t(3i+5)}} \xrightarrow{\mathcal{F}} \frac{1}{(5+i(\omega+3))^2}$

(translația frecvenței)

$(t+2) \cdot e^{-(t+1)(3i+5)} \xrightarrow{\mathcal{F}} \frac{e^{2i\omega}}{(5+i(\omega+3))^2}$

(translația timpului)

Integrale de tipul II-III

(6) $I = \int_0^\infty \frac{x \sin ax}{(x^2+b^2)(x^2+c^2)}\, dx$, $b, c > 0$

Prelungim integrale prin imparitate și atașăm integrala cu cos $\Rightarrow$

$$\left.\begin{aligned} I &= \frac{1}{2} \int_{-\infty}^{\infty} \frac{x \sin ax}{(x^2+b^2)(x^2+c^2)}\, dx \;\Big|\cdot i \\ J &= \int_{-\infty}^{\infty} \frac{x \cos ax}{(x^2+b)(x^2+c^2)}\, dx = 0 \end{aligned}\right\} \Rightarrow$$

$$iI = \underbrace{\int_{\gamma}}_{=0} + iI = \frac{1}{2}\int_{-\infty}^{\infty} \frac{x\cdot e^{iax}}{(x^2+b)(x^2+c^2)}\,dx = \int_{-\infty}^{\infty} \frac{z\cdot e^{iaz}}{(z^2+b)(z^2+c^2)}\,dz$$

Considerăm integrala $D = \int_{\gamma_R} \frac{z\cdot e^{iaz}}{(z^2+b)(z^2+c^2)}\,dz$

unde $\gamma_R = [-R, R] \cup \int_R$, $R > \max(b,c)$.

Funcția $f(z) = \frac{z\cdot e^{iaz}}{(z^2+b^2)(z^2+c^2)}$ este olomorfă cu

poli simpli $\pm ib, \pm ic$.

Cum poli ib și ic sunt în interiorul conturului γ_R, aplicăm teorema reziduurilor și avem

$$D = 2\pi i\,(\operatorname{Rez}(f, ib) + \operatorname{Rez}(f, ic)) =$$

$$\operatorname{Rez}(f, ib) = \lim_{z\to ib} (z-ib)\frac{z\cdot e^{iaz}}{(z-ib)(z+ib)(z^2+c^2)} = \frac{ib\cdot e^{-ab}}{2ib(c^2-b^2)}$$

$$= \frac{e^{-ab}}{2(c^2-b^2)}$$

$$\operatorname{Rez}(f, ic) = \lim_{z\to ic} (z-ic)\frac{z\cdot e^{iaz}}{(z-ic)(z+ic)(z^2+b^2)} = \frac{ic\cdot e^{-ac}}{2ic(-c^2+b^2)}$$

$$= \frac{-e^{-ac}}{2(c^2-b^2)}$$

$$D = 2\pi i\,(\operatorname{Rez}(f, ib), \operatorname{Rez}(f, ic)) = \frac{2\pi i}{2(c^2-b^2)}\left(e^{-ab} - e^{-ac}\right)$$

Pe de altă parte $D = \int_{-R}^{R} \frac{x\cdot e^{iax}}{(x^2+b^2)(x^2+c^2)}\,dx +$

$\int_{\int_R} \frac{x\cdot e^{iax}}{(x^2+b^2)(x^2+c^2)}\,dx$. Trecem la limită $R\to+\infty$

și obținem $\int_{-\infty}^{\infty} \frac{x\cdot e^{-iax}}{(x^2+b^2)(x^2+c^2)}\,dx = \frac{\pi i}{c^2-b^2}\left(e^{-ab} - e^{-ac}\right)$

deoarece conform lemei lui Jordan

$$\lim_{r\to+\infty} \int_{\gamma_R} \frac{x\cdot e^{-iax}}{(x^2+b^2)(x^2+c^2)}\,dx = 0 \quad \left(|xf(x)| = \right.$$

$$\left| \frac{x^2\cdot e^{-iax}}{(z^2+c^2)(z^2+b^2)} \right| < \frac{e^{-y}}{|z^2|} \ (y>0) \Rightarrow \lim_{|z|\to+\infty} |zf(z)| = 0 \Big)$$

Dar $ij + \underbrace{\int_{\gamma}}_{=0} = \frac{1}{2}\int_{-\infty}^{\infty} \text{---}\,dx \Rightarrow$

$$i = \frac{1}{2i}\cdot\frac{\pi i}{c^2-b^2}\left(e^{-ab}-e^{-ac}\right) = \frac{\pi}{2(c^2-b^2)}\cdot\left(e^{-ab}-e^{-ac}\right)$$